Vlastní čísla, vektory – řešení

① Ověřte, že $\lambda = 9$ je vlastní číslo matice $A = \begin{pmatrix} 7 & -2 & -2 \\ -2 & 1 & 4 \\ -2 & 4 & 1 \end{pmatrix}$ a určete odpovídající vlastní vektor.

Stačí určit, zda $|A - \lambda E| = 0$ pro $\lambda = 9$:

$$\begin{vmatrix} 7-9 & -2 & -2 \\ -2 & 1-9 & 4 \\ -2 & 4 & 1-9 \end{vmatrix} = \begin{vmatrix} -2 & -2 & -2 \\ -2 & -8 & 4 \\ -2 & 4 & -8 \end{vmatrix} \; R_2 + (-1)R_1, \; R_3 + (-1)R_1 \; =$$

$$= \begin{vmatrix} -2 & -2 & -2 \\ 0 & -6 & 6 \\ 0 & 6 & -6 \end{vmatrix} \; R_3 + R_2 \; = \underbrace{\begin{vmatrix} -2 & -2 & -2 \\ 0 & -6 & 6 \\ 0 & 0 & 0 \end{vmatrix}}_{(*)} = \boxed{0}$$

$\Rightarrow \lambda = 9$ je vlastní číslo.

Vlastní vektor: $(A - \lambda E)\vec{x} = \vec{o}$ pro $\lambda = 9$:

$$(*) \quad \begin{pmatrix} -2 & -2 & -2 \\ 0 & -6 & 6 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} -2x_1 - 2x_2 - 2x_3 &= 0 \\ -6x_2 + 6x_3 &= 0 \end{aligned}$$

$$\begin{aligned} x_3 &= t \\ -6x_2 + 6x_3 &= 0 \\ 6x_2 &= 6x_3 \\ x_2 = x_3 &= t \\ -2x_1 - 2t - 2t = 0 \;\Rightarrow\; 2x_1 &= -4t \;\Rightarrow\; x_1 = -2t \end{aligned}$$

vlastní vektor: $\boxed{\begin{pmatrix} -2t \\ t \\ t \end{pmatrix}}$

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② Jsou dány body: $A[1,2,2]$, $B[3,1,2]$, $C[5,1,1]$.

a) úhel ABC = úhel vektorů $\overrightarrow{BA}$, $\overrightarrow{BC}$:

$$\overrightarrow{BA} = A - B = (-2,1,0)$$
$$\overrightarrow{BC} = C - B = (2,0,-1)$$
$$\cos\varphi = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{\left\|\overrightarrow{BA}\right\| \cdot \left\|\overrightarrow{BC}\right\|} = \frac{(-2,1,0)\cdot(2,0,-1)}{\sqrt{(-2)^2+1^2+0^2}\cdot\sqrt{2^2+0^2+(-1)^2}} = \frac{-2\cdot 2 + 1\cdot 0 + 0\cdot(-1)}{\sqrt{5}\cdot\sqrt{5}} = -\frac{4}{5}$$

$$\sphericalangle ABC = \varphi = \arccos\left(-\frac{4}{5}\right)$$

b) obsah $\triangle ABC$**:**

$$S_\triangle = \frac{1}{2}\left\|\overrightarrow{BA} \times \overrightarrow{BC}\right\| = \frac{1}{2}\sqrt{(-1)^2+(-2)^2+(-2)^2} = \frac{1}{2}\cdot\sqrt{9} = \boxed{\frac{3}{2}}$$

$$\overrightarrow{BA} \times \overrightarrow{BC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -2 & 1 & 0 \\ 2 & 0 & -1 \end{vmatrix} = \vec{i}\cdot(-1-0) - \vec{j}\cdot(2-0) + \vec{k}\cdot(0-2) = (-1,-2,-2)$$

c) délka úsečky AB**:**

$$|AB| = \left\|\overrightarrow{AB}\right\| = \left\|\overrightarrow{BA}\right\| = \sqrt{(-2)^2+1^2+0^2} = \boxed{\sqrt{5}}$$